



GMR Institute of Technology
An Autonomous Institute Affiliated to JNTU-GV



STEPCONE 2025

TECHNICAL PAPER PRESENTATION



TECHNICAL PAPER PRESENTATION - 2025

The STEPCONE platform aims to showcase empirical research outcomes across diverse engineering domains, emphasizing the identification of research gaps and advancements in emerging areas of engineering and technology. The platform also encourages the examination of current scientific contributions relevant to engineering research. Research communication, encompassing both novel perspectives on traditional domains and cutting-edge technologies, is highly encouraged.

We invite full papers in the prescribed format from enthusiastic student researchers. Submissions are accepted either as a single-author contribution or as a collaborative effort with a maximum of two authors.

We look forward to receiving innovative research contributions that contribute to the dialogue on engineering advancements and further enrich the STEPCONE platform with valuable insights.

GUIDELINES FOR PAPER PRESENTATION & SUBMISSION

Submission Guidelines:

1. **Abstract Length:** Please submit an abstract of 100-150 words, providing a concise overview of your research paper.
2. **Submission Deadline:** Abstract submission deadline is **February 8, 2025**.
3. **Intimation of Acceptance:** Notifications of acceptance will be sent **on or before February 15, 2025**.

Full Paper Submission:

4. **Selected Candidates:** Shortlisted candidates will be notified to submit the full paper.
5. **Full Paper Length:** The full paper should be between 2000-5000 words, formatted in **IEEE citation and reference style**. The paper should be submitted in hard copy to the organizers and the soft copy should be submitted at

https://docs.google.com/forms/d/e/1FAIpQLSe0aD36FIU-8IJhCaOh80QtNOmsUrkuertzASBqBFtFQksF1-g/viewform?usp=sf_link

6. Formatting Requirements:

- **Font:** Times New Roman.
- **Font Size:** 12 (regular), 14 (bold for headings).
- **Content Order:**

Paper contents should be arranged in the following order but can be modified if discipline/nature of work demands:

- Title
- Abstract
- Keywords
- Introduction
- Materials & Methods
- Results & Discussions
- Conclusion
- References

7. **Submission Deadline:** Full papers must be submitted by **February 18, 2025**, for consideration.

8. Author Details:

- Include:
 - Serial order for all authors.
 - Current Year of Study.
 - Branch.
 - Institute Name and Address.
 - Email ID.

- Phone Number.

9. **Corresponding Author:** One author must submit on behalf of the group and will serve as the point of contact and be treated as corresponding author.

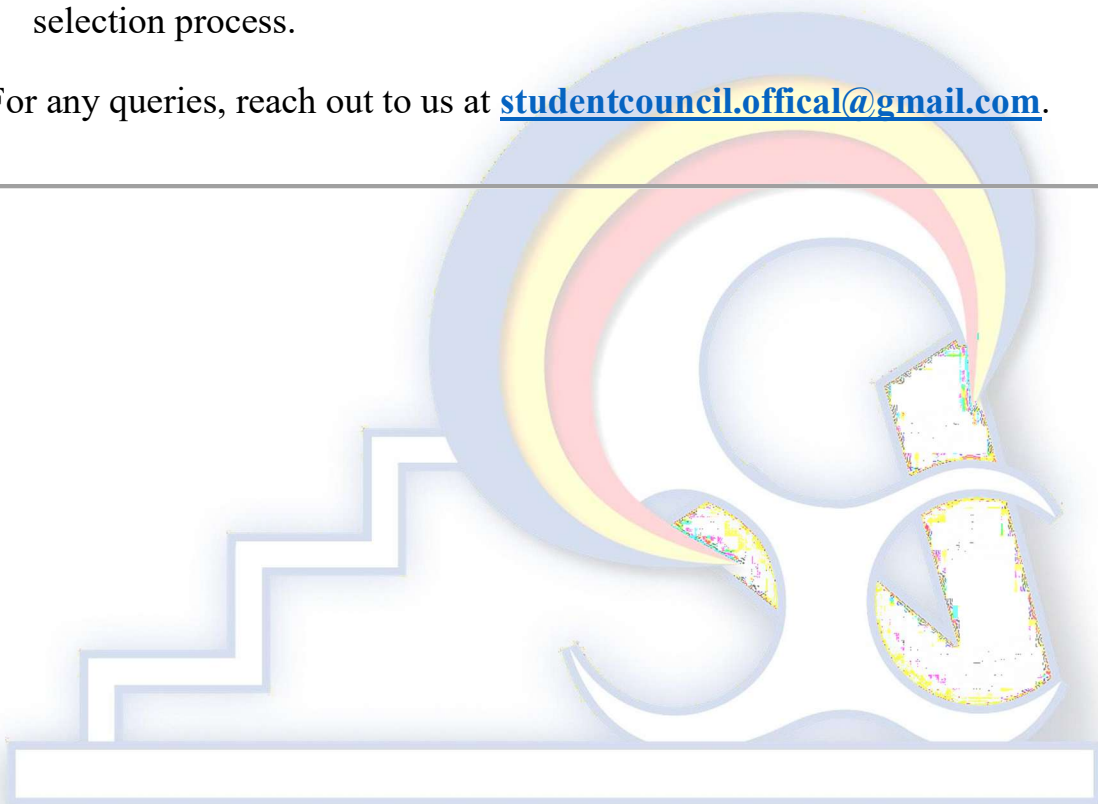
Presentation Details:

10.**Duration:** Each participant gets **10 minutes** for their presentation.

11.**File Submission:** Presentation file (named with Paper ID) must be submitted by **February 20, 2025**.

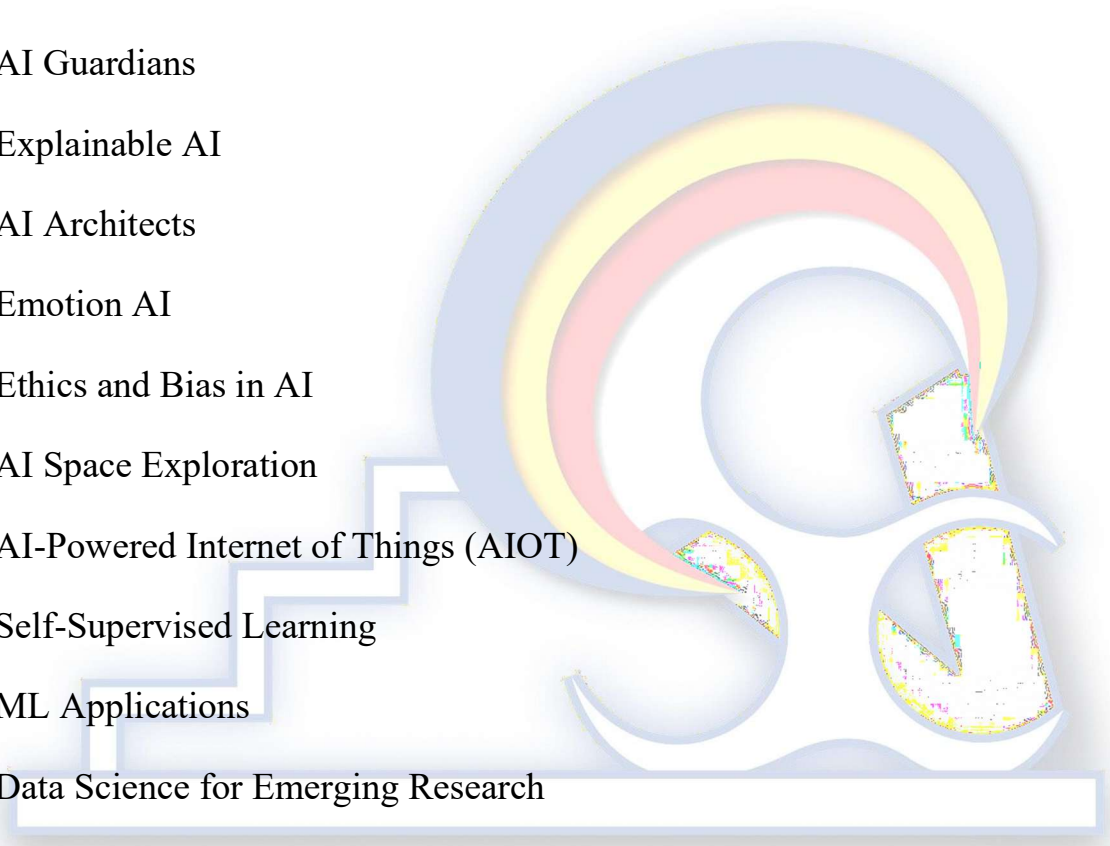
12.**Schedule:** Presentation schedules will be communicated after the abstract selection process.

For any queries, reach out to us at studentcouncil.offical@gmail.com.



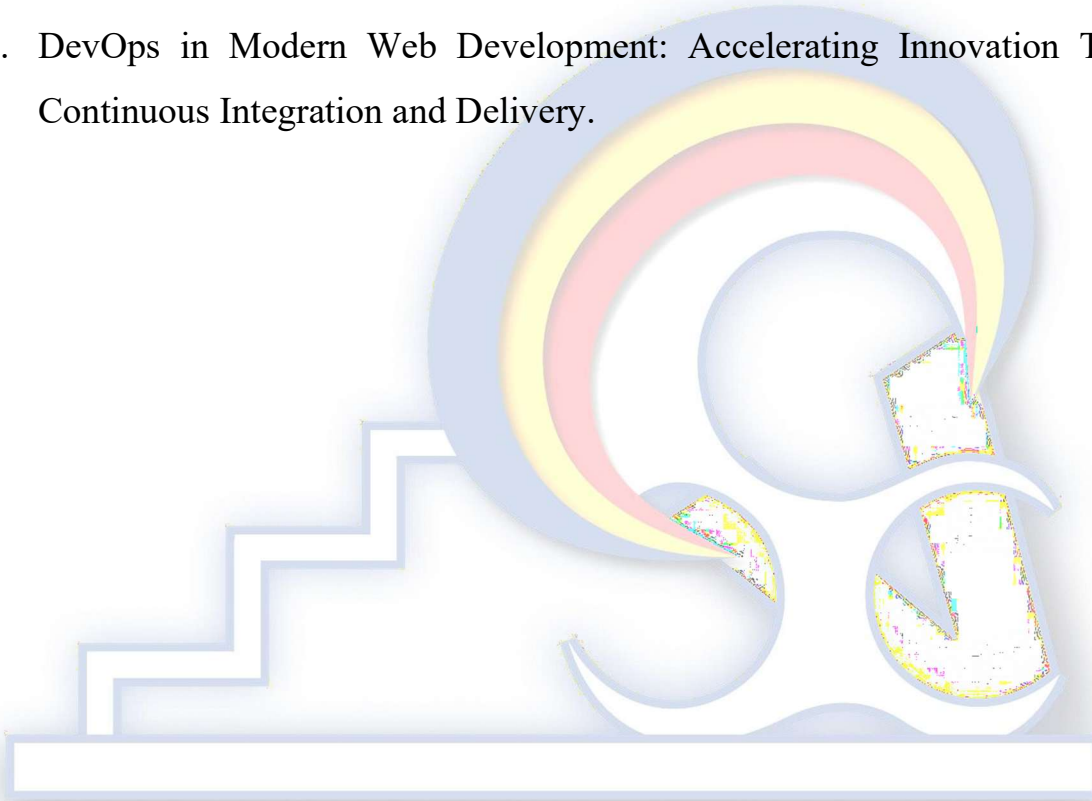
TECHNICAL PAPER PRESENTATION THEMES

2.1. Computer Science and Engineering, Information Technology, Artificial intelligence and machine learning, Artificial intelligence, and Data Science

1. The GenAI-LLM Nexus
 2. Smart Care AI
 3. AI Guardians
 4. Explainable AI
 5. AI Architects
 6. Emotion AI
 7. Ethics and Bias in AI
 8. AI Space Exploration
 9. AI-Powered Internet of Things (AIOT)
 10. Self-Supervised Learning
 11. ML Applications
 12. Data Science for Emerging Research
 13. Hardware Models Using IoT
 14. Data Analytics Using IoT
 15. AI for Engineering Applications
 16. Blockchain Technology for Cybersecurity Enhancement: Applications, Challenges, and Future Perspectives
 17. Artificial Intelligence in Cybersecurity: A Double-Edged Sword for Defense
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and Exploitation

18. Reinforcement Learning in Machine Learning: Applications, Techniques, and Real-World Impacts
19. Federated Learning in Machine Learning: Privacy-Preserving Distributed Data Training and Applications
20. DevOps in Modern Web Development: Accelerating Innovation Through Continuous Integration and Delivery.



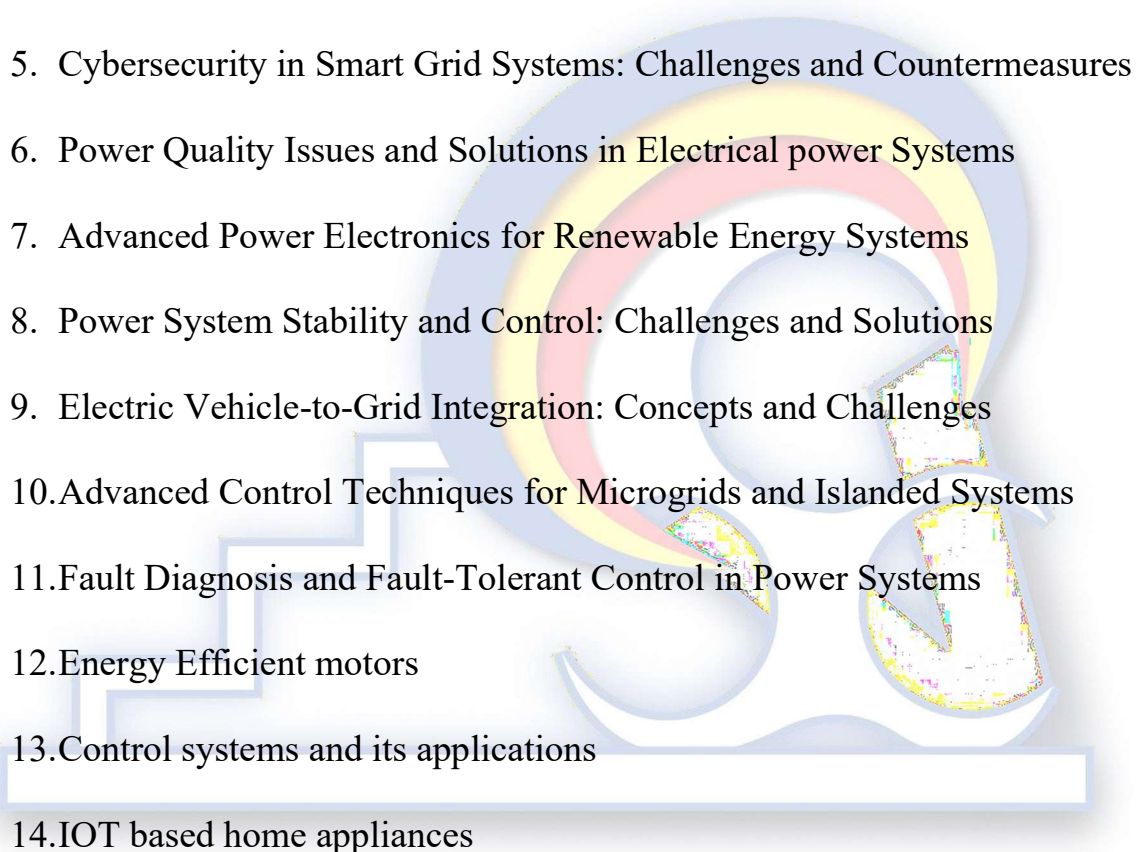
2.2 Civil Engineering

1. Integration of AI and ML in Civil Engineering
 2. 3D Printing in Building Construction
 3. Big Data Analytics for Predictive Maintenance in Civil Engineering
 4. IoT in Civil Engineering
 5. Virtual Reality in Construction Planning
 6. Sustainable Materials for Green Building Construction
 7. Autonomous Vehicles and their Impact on Urban Infrastructure
 8. Advanced Technologies to Enhance the Structural Resilience Against Earthquakes
 9. Sustainable Structural Maintenance and Rehabilitation
 10. Intelligent Transport Systems
 11. Use of Nanomaterials in Civil Engineering
 12. Soil Stabilization using sustainable materials
 13. Sustainable Water Management Systems
 14. Applications of Remote Sensing and GIS in Civil Engineering
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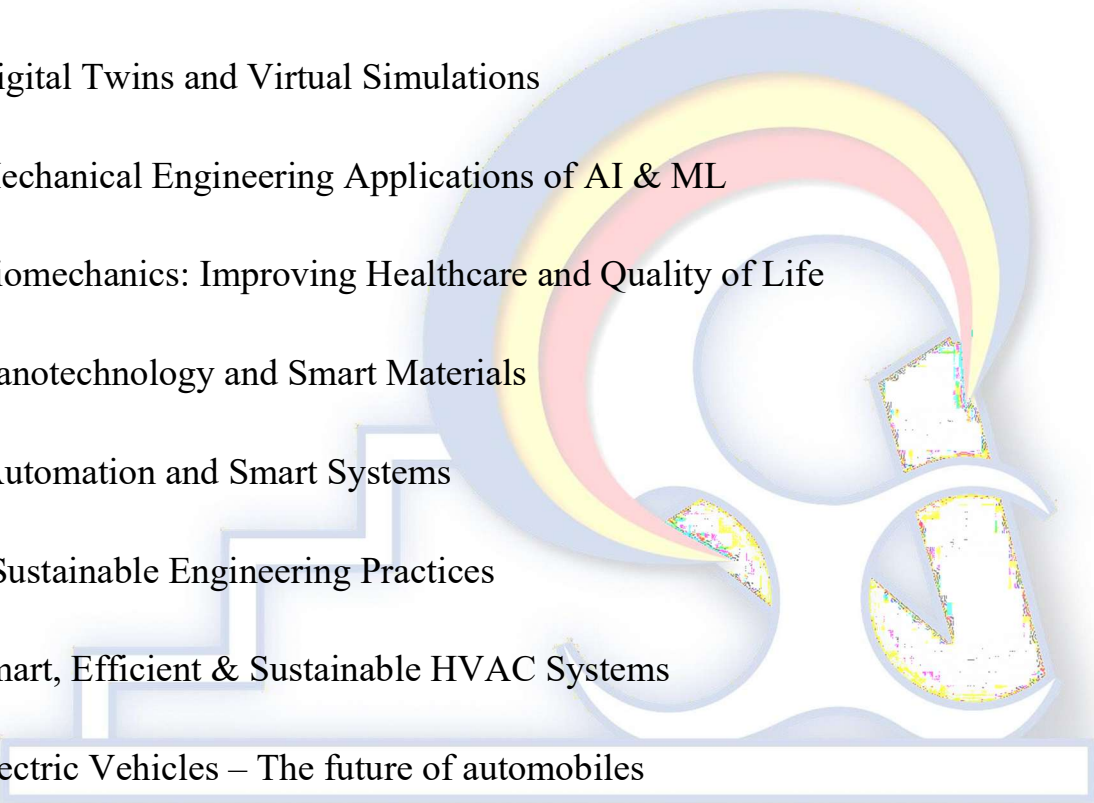
2.3 Electronics and Communication Engineering

1. 5G and Beyond: The Future of Wireless Communication
 2. Quantum Computing and its Applications in Communication
 3. Underwater Wireless Communication: Challenges and Solutions
 4. Li-Fi Technology: The Future of Data Transmission
 5. Low-Power VLSI Design for IoT Devices
 6. MEMS and Nano-electronics in Healthcare Applications
 7. Deep Learning for Image and Video Processing
 8. Speech Recognition Systems: Advances and Applications
 9. Green Communication Networks: Challenges and Solutions
 10. Solar-Powered Communication Systems for Remote Areas
 11. Cryptography in Secure Communication Systems
 12. Cyber Security in Embedded Systems
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2.4 Electrical and Electronics Engineering

1. Renewable Energy Integration into Power Grids
 2. Artificial Intelligence Applications in Power Systems
 3. Emerging Trends in Smart Grids and Energy Management
 4. Advanced Control Techniques for Power Electronics Converters
 5. Cybersecurity in Smart Grid Systems: Challenges and Countermeasures
 6. Power Quality Issues and Solutions in Electrical power Systems
 7. Advanced Power Electronics for Renewable Energy Systems
 8. Power System Stability and Control: Challenges and Solutions
 9. Electric Vehicle-to-Grid Integration: Concepts and Challenges
 10. Advanced Control Techniques for Microgrids and Islanded Systems
 11. Fault Diagnosis and Fault-Tolerant Control in Power Systems
 12. Energy Efficient motors
 13. Control systems and its applications
 14. IOT based home appliances
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2.4 Mechanical Engineering

1. Green Technologies for a Sustainable Future
 2. Digitalization and Industry 4.0
 3. Additive Manufacturing through 3D Printing
 4. Sustainable Business Practices in Modern Corporations
 5. Digital Twins and Virtual Simulations
 6. Mechanical Engineering Applications of AI & ML
 7. Biomechanics: Improving Healthcare and Quality of Life
 8. Nanotechnology and Smart Materials
 9. Automation and Smart Systems
 10. Sustainable Engineering Practices
 1. Smart, Efficient & Sustainable HVAC Systems
 13. Electric Vehicles – The future of automobiles
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Topics related to contemporary engineering fields are also considered